

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Austin school **\$2.12M** over the next 20 years.

That is 62.4% less than the reactive cost, or \$922K in today's dollars. It compares PaxSeal planned maintenance with fixing problems only after they leak, using Austin costs and storm history.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Austin	K-12	120,000 sq ft	Mod bit	8 yrs Good	\$1.05M

PAXSEAL NET COST

\$1.27M

\$2.44M spent, less \$1.16M roof life left

REACTIVE NET COST

\$3.39M

\$3.56M spent, less \$172K roof life left

EMERGENCY COSTS CUT

\$1.78M

Unbudgeted repairs and storm damage avoided; \$4.80 back per fee dollar

NEW ROOF PRICE

2× in 15 yrs

\$1.05M today, \$2.58M by year 20

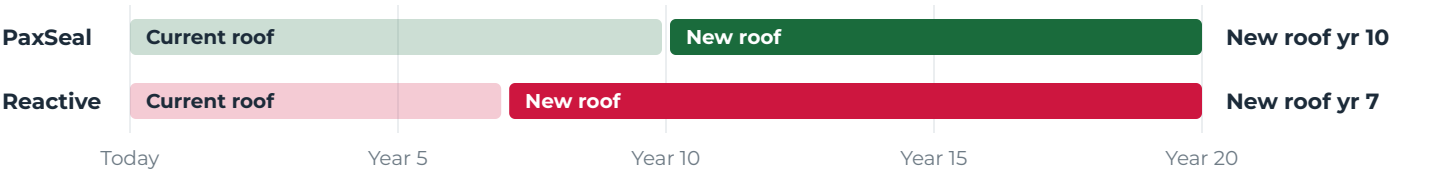
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$441K	—	-\$441K
Repairs paid by owner	\$252K	\$1.98M	+\$1.73M
Expected storm damage	\$97K	\$145K	+\$48K
Roof replacements	\$1.65M	\$1.44M	-\$208K
Credit for roof life left	-\$1.16M	-\$172K	+\$989K
Net 20-year cost	\$1.27M	\$3.39M	+\$2.12M

What this means

- 1

A later replacement. Maintenance moves the next new roof from year 7 to year 10, so fewer roofs are bought over the period.
- 2

Smaller repairs. Problems found on a scheduled visit avoid emergency work, which costs about 3× as much. That saves \$1.73M in repairs.
- 3

A predictable budget line. PaxSeal turns unplanned emergency repairs into a fixed annual cost the board can plan for, and keeps leaks from closing classrooms.
- 4

The program pays for itself. Each \$1 of PaxSeal fees returns about \$4.80 in avoided cost.

Local risk profile

Travis County, TX (48453)

Hail and tornado

~9 days with 2"+ hail somewhere in Travis County per 20 years (NOAA 1996–2025). Expected storm damage to this roof over 20 years is **\$97K** if maintained and **\$145K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Very High	■■■■■
Strong wind	High	■■■■■
Tornado	Very High	■■■■■
Hurricane	Low	■■■■■
Winter weather	Moderate	■■■■■
Ice storm	High	■■■■■
FEMA National Risk Index rating		

Key assumptions

Confidence reflects source strength. Values will be refined as research continues.

ASSUMPTION	VALUE	BASIS	CONFIDENCE
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs from NRC Canada roof study and manufacturer warranty terms	MEDIUM
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025	HIGH
Replacement cost	\$8.75/sq ft	Median of other PAX markets until local data is added	LOW
Reactive repair cost	3× planned	Industry estimates; no primary study found	LOW
Storm damage	0.28%/yr	Of roof cost. FEMA NRI expected annual building loss from hail, wind, tornado, hurricane and winter weather	MEDIUM
PaxSeal fee	\$16,400/yr	PAX program estimate for this roof size. Your proposal will state the actual fee	ESTIMATE
Leak consequences	1.30× warehouse	Scales reactive repair costs. Order supported by school-closure, hospital infection-control and insurer water-damage evidence; exact values are estimates	LOW

Estimates, not a quote. Projections use Austin market research and industry data, in nominal dollars; actual costs will differ. The PaxSeal fee is an estimate and your proposal will state the actual price. Net cost credits the roof life remaining at year 20.